UTKARSH MISHRA

FIELD & BEHAVIOURAL RESEARCHER | PROGRAMME EVALUATION | HUMAN-AI INTERACTION

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Research Profile

Field and behavioural researcher with experience designing and implementing large-scale studies and programme evaluations in India. Experienced in survey design, field-team coordination, primary data collection, mixed-methods research, and quantitative analysis in R and Python, with additional research experience in Human-AI Interaction and AI for education.

Research interests: AI in education and public services, field evaluation of AI systems, Human-AI Interaction, behavioural science, and programme evaluation.

Education

Indian Institute of Technology, Delhi (IIT Delhi) 2023 – 2025
M.Sc. in Cognitive Science *GPA: 7.5/10*
Relevant Coursework: Advanced Data Analysis for Behavioural Research using R, Advanced Experimental Methods, Advanced Computational Methods, Research Methods, Scientific Writing, Design for User Experience, Design for Usability.

Jamia Millia Islamia, New Delhi 2020 – 2023
B.A. in Economics & Psychology *GPA: 8.33/10*
Relevant Coursework: Microeconomics, Macroeconomics, Money, International Trade, Indian Economy, Theories of Public Administration, Indian Public Administration.

Research Experience

Research Associate, IIM Ahmedabad
Selected Field Research & Programme Evaluation

Large-Scale Field Evaluation, Rural Development Programme (MoRD, Government of India)
Collaborating PIs: Prof. Jeevant Rampal, Prof. Anindya

- **Co-designed and implemented** a large-scale mixed-methods evaluation study of a Government of India rural development programme, defining outcome indicators, contributing to sampling strategy, and conducting stakeholder consultations across administrative levels.
- Designed, coded, translated, and **piloted beneficiary and field-cadre survey instruments in Qualtrics**, using literature review, implementation analysis, and continuous stakeholder inputs.
- Led **primary data collection across three states** ($n > 6{,}000$), recruiting, training, and managing **field teams of 30+ researchers per district** and coordinating field operations across study sites.
- Cleaned, organised, and analysed survey data in **R and Python** and created composite outcome indices. Performed regression, balance checks, sentiment analysis for further analyses.
- Conducted **field interview** in an additional state for **case-study development**, analysed qualitative data thematically and co-authored the case study. Also co-authored the final report and contributed to presentations to the **Ministry of Rural Development**, informing programme revisions.

Ex-ante Impact Assessment, India's Olympic Bid 2036
In collaboration with EY and Gujarat Sports Infrastructure Development (GSID), Govt. of Gujarat.
Collaborating PIs: Prof. Jeevant Rampal, Prof. Anindya, Prof. Anish, Prof. Amit

- Designed and implemented a **large-scale public perception study** of approximately **6,000 respondents** examining attitudes and expectations toward India's proposed 2036 Olympic bid.
- Managed **survey design, Qualtrics implementation, primary data collection, sampling strategy and quantitative analysis**, integrating recorded structured responses with open-ended

feedback to identify population-level patterns of positive and negative perceptions and attitude.
- Applied **sentiment analysis and topic-modelling in R and Python** to analyse open-ended responses and identify recurring positive and negative perceptions across the respondent pool.
- Assisted in drafting the study plan and making a **pre-registration** on **Open Science Forum.**
- Used secondary data to make socio-economic projections on relevant indicators for the year 2036.

Research Assistant, Department of Design, IIT Delhi
Psychometric Instrument Development *Supervised by Prof. Jyoti*

- Assisted in the designed of a **quantitative psychometric instrument** measuring children's subjective experience of enjoyment during educational play (ages 3–6).
- Assisted in conducting **expert-panel review and cognitive interviews** to refine item wording and assess comprehension across respondent backgrounds.
- Administered the instrument to **130 participants** in Delhi and validated its structure using **EFA, CFA, and internal consistency analysis** (Cronbach's α).

Selected Human-AI & Behavioural Research

Master's Thesis, IIT Delhi
AI for Digital Learning, Human-AI Interaction *Supervised by Prof. Jyoti, Head, Dept. of Design*

- Investigated how **AI systems can support adaptive digital learning** by identifying characteristics of readable learning content and generating adaptive content in Large Language Models.
- Developed an attention-based **multimodal AI model** integrating text and audio for emotion recognition in conversational learning contexts; addressed class imbalance through ensemble learning.
- Developed an AI pipeline using a **fine-tuned Gemma model** to generate adaptive learning content.

Research Project
Perceived AI Assistance & Human Behaviour

- Investigated psychological underpinnings of how **perceived AI assistance** influences user experience and task performance by reanalysing data from a published Human-AI Interaction study.
- Applied **Bayesian statistical modelling and inference in Python** to estimate individual-level psychological parameters and quantify uncertainty in behavioural effects.
- Findings accepted for poster presentation at the **Annual Conference of Cognitive Science**.

Human-Computer Interaction Researcher, Vibe AI
Human-AI Interaction and User Research

- Conducted generative user research using **focus groups, semi-structured interviews, and surveys** to investigate user needs, behavioural patterns, and pain points in AI product interactions.
- Evaluated AI product prototypes with **100+ users**, combining behavioural measures with post-use interviews to identify interaction barriers, decision triggers, and sources of UX friction.
- Synthesized qualitative and behavioural evidence into actionable recommendations and collaborated with product teams to translate findings into human-centred improvements to AI interactions.

Additional Research Experience

Research Intern, Department of Psychology, King's College London
Bayesian Computational Modelling of Pain Perception *Supervised by Dr. Devin Terhune; remote collaboration*

- Reanalysed experimental data using competing **Bayesian computational models** and MATLAB's Variational Bayesian Analysis (VBA) toolbox to investigate mechanisms of placebo hypoalgesia.
- Compared competing state-space models, achieving **84% predictive accuracy**, and estimated model parameters using Variational Bayesian inference.
- Developed the confirmatory study design, research question, hypotheses, analysis plan, and **OSF**

preregistration for subsequent analyses.

Research Project
Zomato Consumer & Usability Research

- Conducted task-based behavioural research examining navigation, information-seeking behaviour, decision triggers, and interaction friction in restaurant discovery and booking.
- Evaluated redesigned interface concepts through **usability testing and A/B testing**, reducing task-completion time by **18%** and increasing task-completion rate by **15%**.

Fellowships

- Selected as a Research Fellow at I-Hub Foundation for Cobotics (IHFC), **Technology and Innovation Hub (TIH)**, IIT Delhi set up under the **Department of Science & Technology**'s NM-ICPS mission. Worked with the **Education, Research and Training vertical.**

Selected Publications & Conference Presentations

- Mishra, U., Parsons, R., Wiech, K. & Terhune, D. (2026). *Investigating the link between trait responsiveness to verbal suggestion and the placebo effect within the predictive processing framework.* Accepted for Poster Presentation, **13th Annual Conference of Cognitive Science (ACCS)**.(Could not attend)
- Mishra, U. (2026). *Impact of Perceived AI-based Assistance on User Experience and Task Performance.* Accepted for Poster Presentation, **13th Annual Conference of Cognitive Science (ACCS)**.(Could not attend)

Research Methods & Technical Skills

- **Field Research:** Survey design, mixed-methods research, primary data collection, field-team coordination, programme evaluation, stakeholder research, qualitative interviews, cognitive interviewing.
- **Quantitative Methods:** R, Python (pandas, numpy, scipy, scikit-learn), regression analysis, EFA/CFA, Bayesian statistical modelling, sentiment analysis, statistical inference.
- **Survey & Data Collection:** Qualtrics, survey instrument development, survey translation and piloting, behavioural observation, usability testing.
- **Human-AI Research:** Human-AI Interaction, AI-assisted learning, NLP, LLMs, multimodal ML, behavioural research, user research.
- **Other Tools:** MATLAB, Psychtoolbox, Git/GitHub, LaTeX, MS Excel.

Languages

Hindi: Native proficiency **English:** Full professional proficiency